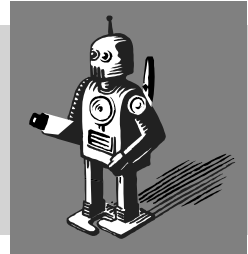


5

Transformations



For the purpose of controlling a robot, it is necessary to know the relationships between the joints' motion (input) and the end-effector motion (output), because the joint motions control the end-effector's movements. Thus, the study of kinematics is important, where transformations between the coordinate frames attached to different links of the robot need to be performed.

5.1 ROBOT ARCHITECTURE

A robot under study is made up of several links connected serially by joints. The robot's Degree Of Freedom (DOF) depends on the number of links and joints, their types, and the kinematic chain of the robot.

5.1.1 Links and Joints

The individual bodies that make up a robot are called *links*. Here, unless otherwise stated, all links are assumed to be rigid, i.e., the distance between any two points within the body does not change while it is moving. A rigid body in the three-dimensional Cartesian space has six DOF. This implies that the position of the body can be described by three translational, and the orientation by three rotational coordinates. For convenience, certain nonrigid bodies, such as chains, cables, or belts, which when serve the same function as rigid bodies, may be considered as links. From the kinematic point of view, two or more members when connected together without any relative motion between them are considered a single rigid link. For example, an assembly of two gears connected by a common shaft is treated as one link.

Links of a robot are coupled by kinematic pairs or joints. A *joint* couples two links and provides physical constraints on the relative motion between the links. It is not a physical entity but just a concept that allows one to specify how one link moves with respect to another one. For example, a hinge joint of a door allows it to move relative to the fixed wall about an axis. No other motion is possible. The type of relative motion permitted by a joint is governed by the form of the contact surface between the members, which can be a surface, a line, or a point. Accordingly, they are termed as either *lower* or *higher* pair joint. If two mating links are in surface contact, the joint is referred to as a lower pair joint. On the contrary, if the links are in line or point contact, the joint is called higher pair joint. As per the definition, the hinge joint of the door is a lower pair joint, whereas a ball rolling on a plane makes a higher pair joint. Frequently used lower and higher pairs are listed in Table 5.1.

1. Revolute Joint, R A revolute joint, also known as a *turning pair* or a *hinge* or a *pin joint*, permits two paired links to rotate with respect to each other about the axis of the joint, say, Z , as shown in Fig. 5.1. Hence, a revolute joint imposes five constraints, i.e., it prohibits one of the links to translate with respect to the other one along the three perpendicular axes, X , Y , Z , along with the rotation about two axes, X and Y . This joint has one degree of freedom (DOF).

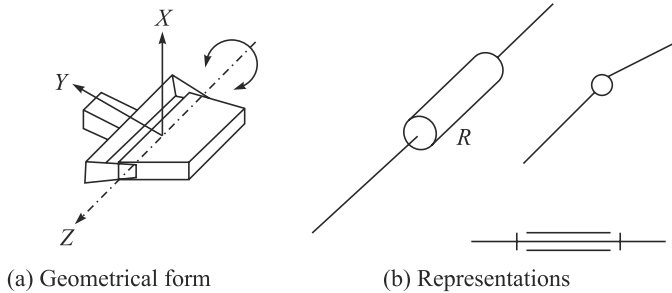


Fig. 5.1 A revolute joint

2. Prismatic Joint, P A *prismatic joint* or a *sliding pair* allows two paired links to slide with respect to each other along its axis, as shown in Fig. 5.2. It also imposes five constraints and, hence, has one DOF.

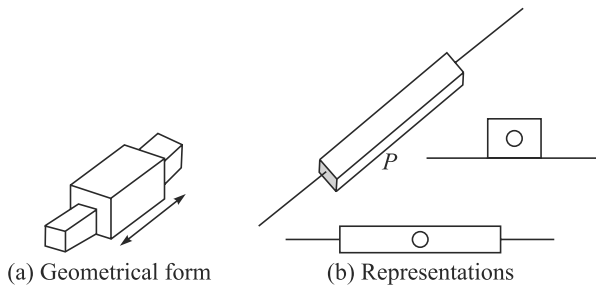


Fig. 5.2 A prismatic joint

3. Helical Joint, H As shown in Fig. 5.3, a helical joint allows two paired links to rotate about and translate at the same time along the axis of the joint. The translation is, however, not independent. It is related to the rotation by the pitch of the screw. Thus, the helical joint also has five constraints, and accordingly one DOF.

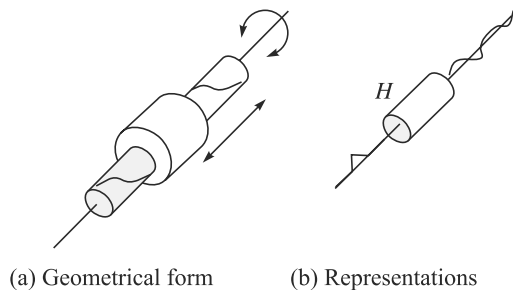


Fig. 5.3 A helical joint

4. Cylindrical Joint, C It permits rotation about, and independent translation along, the axis of the joint, as shown in Fig. 5.4. Hence, a cylindrical joint imposes four constraints on the paired links, and has two DOF.

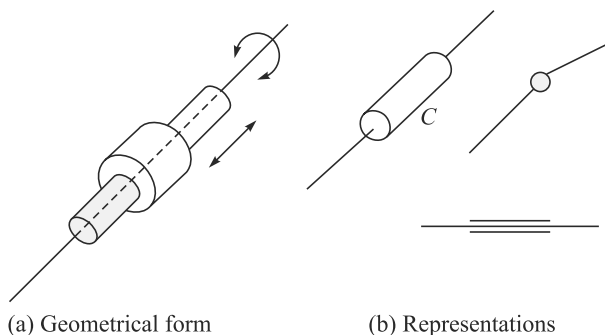


Fig. 5.4 A cylindrical joint

5. Spherical Joint, S It allows one of the coupled links to rotate freely in all possible orientation with respect to the other one about the center of a sphere. No relative translation is permitted. Hence, it imposes three constraints and has three DOF.

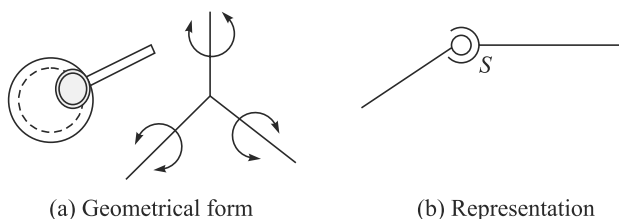


Fig. 5.5 A spherical joint

6. Planar Joint, L This three DOF joint allows two translations along the two independent axes of the plane of contact and one rotation about the axis normal to the plane, Fig. 5.6.

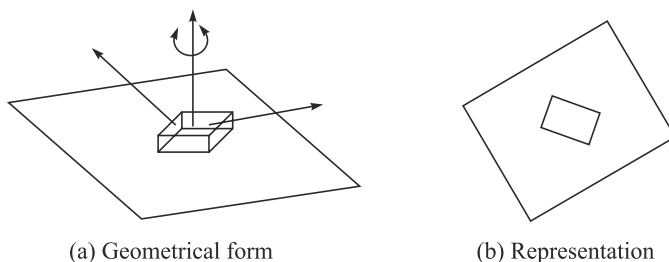
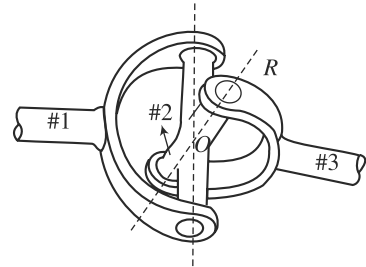


Fig. 5.6 A planar joint

Table 5.1 Lower pair joints

Name	Symbol	Geometric Form and Representations	DOF	Common Surface
Revolute	R	Fig. 5.1	1	Cylinder
Prismatic	P	Fig. 5.2	1	Prism
Helical	H	Fig. 5.3	1	Screw
Cylindrical	C	Fig. 5.4	2	Cylinder
Spherical	S	Fig. 5.5	3	Sphere
Planar	L	Fig. 5.6	3	Plane

Table 5.1 summarizes the basic lower pair joints, where all the joints have surface contact between the interconnecting links. Another commonly used lower pair joint in robotics is the two DOF *universal joint*, as shown in Fig. 5.7. This is the combination of two intersecting revolute joints. Examples of higher pair joints in robots are *gears* and *cams with roller followers*, where they make line contacts.

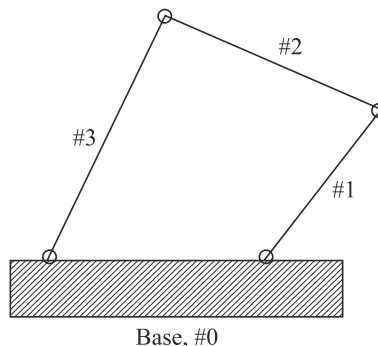

Fig. 5.7 A universal joint

5.1.2 Kinematic Chain

A *kinematic chain* is a series of links connected by joints. When each and every link in a kinematic chain is coupled to at most two other links, the chain is referred to as *simple kinematic chain*. A simple kinematic chain can be either *closed* or *open*. It is closed if each and every link is coupled to two other links as shown in Fig. 5.8. A kinematic chain is open if it contains exactly two links, namely, the end ones that are coupled to only one link. A robotic manipulator shown in Fig. 5.9 falls in this category.

Mechanism vs. Manipulator

Whereas a mechanism can be open- and closed-chain mechanical system; a manipulator is treated here as an open-chain system.


Fig. 5.8 A four-bar mechanism

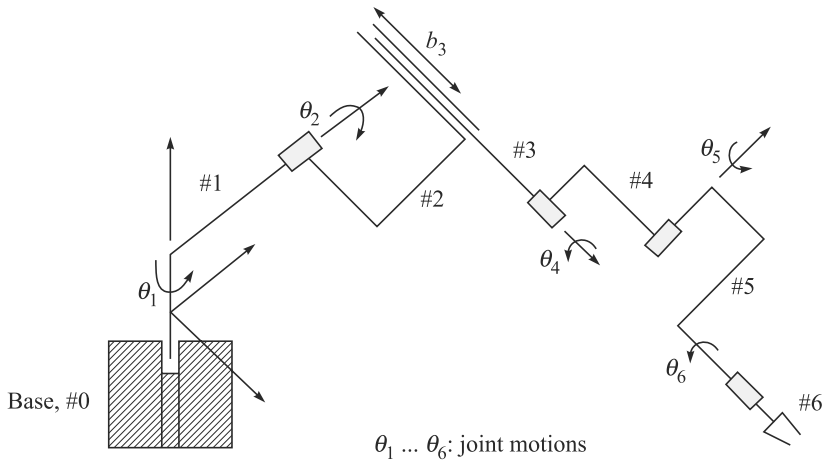


Fig. 5.9 A robot manipulator

5.1.3 Degree of Freedom

Formally, the degree of freedom (DOF) of a mechanical system is defined as the number of independent coordinates or minimum coordinates required to fully describe its pose or configuration. Thus, a rigid body moving in the three-dimensional Cartesian space has six DOF, three each for position and orientation. Several methodologies exist to determine the DOF. One such method was given by Grubler in 1917 for planar mechanisms, which was later generalized by Kutzbach in 1929 for spatial mechanisms. Together they are known as Grubler Kutzbach criterion, which is derived below:

Assume,

s : dimension of working space (for planar, $s = 3$; spatial, $s = 6$)

r : number of rigid bodies or links in the system

p : number of kinematic pairs or joints in the system

c_i : number of constraints imposed by each joint

c : total number of constraints imposed by p joints

n_i : relative degree of freedom of each joint

n : degree of freedom (DOF) of the whole system

The DOF of the whole system n is then determined from the total DOF associated with all the moving links minus the total number of constraints imposed by all the joints, i.e.,

$$n = s(r - 1) - c, \text{ where } c \equiv \sum_{i=1}^p c_i \quad (5.1)$$

Note that -1 in Eq. (5.1) corresponds to the fixed link of the system which has zero DOF. Moreover, for the coupled links, the sum of the number of constraints imposed by each joint, c_i , and the relative degree of freedom permitted by that joint, n_i , is

equal to the dimension of the working space, s , i.e., $s = c_i + n_i$. Hence, the total number of constraints imposed by all the joints can be re-written as

$$c = \sum_{i=1}^p c_i = \sum_{i=1}^p (s - n_i) = sp - \sum_{i=1}^p n_i \quad (5.2)$$

Upon substitution of Eq. (5.2) into Eq. (5.1), one obtains the DOF of the system, n , as

$$n = s(r - p - 1) + \sum_{i=1}^p n_i \quad (5.3a)$$

which is the well-known Grubler Kutzbach criterion for the determination of a system's DOF. A simple and easy-to-use form of Eq. (5.3a) for the planar and spatial motions of a system consisting of only one DOF kinematic pairs, e.g., revolute or prismatic, can be given by

$$n = 3(r - 1) - 2p: \text{For planar}; n = 6(r - 1) - 5p: \text{For spatial} \quad (5.3b)$$

Note that the link dimensions are not entered in the formula, which sometimes affect the DOF of the mechanism due to its geometrical arrangement of the links.

Example 5.1 Four-bar Mechanism

Figure 5.8 shows a planar four-bar mechanism whose links, including the ground, are connected in a loop by four revolute joints. Here, $s = 3$, $r = 4$, and $p = 4$. Equation (5.3a) gives

$$n = 3(4 - 4 - 1) + (1 + 1 + 1 + 1) = 1 \quad (5.4)$$

Hence, the planar four-bar linkage has one DOF. One can easily verify the result of Eq. (5.4) using the expression of Eq. (5.3b) for planar motion too.

Example 5.2 A Robot Manipulator

Figure 5.9 shows the schematic diagram of a Stanford robot (Angeles, 2003). Its DOF can be determined using Eq. (5.3a) as $s = 6$, $r = 7$ (including the fixed base), $p = 6$. Then, from Eq. (5.3a), the DOF of the robot is

$$n = 6(7 - 6 - 1) + 6 \times 1 = 6 \quad (5.5)$$

One can easily verify the result of Eq. (5.5) using the expression of Eq. (5.3b) for spatial motion too.

Example 5.3 Five-bar Mechanism

Figure 5.10 shows a planar five-bar mechanism. There are five links connected in a loop by five revolute joints. Hence, $s = 3$, $r = 5$ (including the base), and $p = 5$. Equation (5.3a) yields

$$n = 3(5 - 5 - 1) + 5 \times 1 = 2 \quad (5.6)$$

The planar five-bar mechanism has two DOF, which is often used as a robot manipulator also.

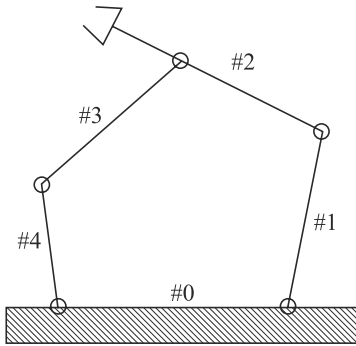


Fig. 5.10 A five-bar mechanism

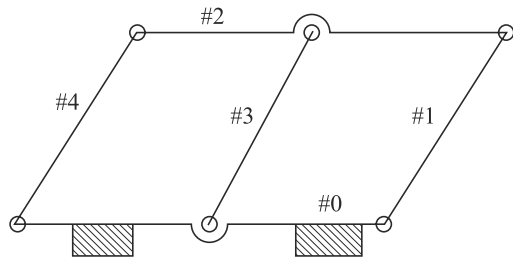


Fig. 5.11 A double parallelogram mechanism

Example 5.4 A Double Parallelogram Mechanism

Figure 5.11 shows a double parallelogram mechanism, where $s = 3$, $r = 5$ (including the fixed base), $p = 6$. Then, Eq. (5.3a), gives

$$n = 3(5 - 6 - 1) + 6 \times 1 = 0 \quad (5.7)$$

which interprets the system as a structure with zero DOF. This is not true, as the mechanism is known to have one DOF. Actually, the link #5 has length equal to that of link #2. Hence, #5 is kinematically redundant and can be removed without destroying the basic kinematic chain and subsequently the input-output behavior of the mechanism.

In general, if the Grubler–Kutzbach criterion, Eq. (5.3), yields $n > 0$, the mechanism has n DOF. If the criterion yields $n = 0$, it is a structure with zero DOF, and if $n < 0$, the system is a statically indeterminate structure with redundant constraints. Exceptions, however, occur as the one shown for Fig. 5.11 which does not obey the Grubler–Kutzbach criterion. For such systems, different mobility criterion, e.g., loop mobility criterion, as described by Tsai (1999), etc., can be employed to determine their DOF.

5.2 POSE OF A RIGID BODY

Rigid-body motion in the three-dimensional Cartesian space comprises of translation and rotation. Whereas translation is defined using three Cartesian coordinates, the rotation needs three angular coordinates. Hence, the rigid-body motion can be defined completely using six coordinates. In the study of the kinematics of robot manipulators, one constantly deals with the position and orientation of several bodies in space. The bodies of interest include the links of the manipulator, tools, and workpiece. To identify the position and orientation of a body, i.e., its *pose* or *configuration*, a fixed reference coordinate system is established, which is called the *fixed frame*. Next, a Cartesian coordinate system attached to the moving body is employed to describe its pose.

Pose vs. Configuration

They mean the same in this book, i.e., the position of a point on a rigid body and its orientation.